

Initiating Search

November 4, 2025, 2:48 PM

• Search:

CAS Lexicon Search:

Julia-Kocienski olefination - Preferred Concept

Julia-Kocienski olefination - Broader Concepts (1)

Julia-Kocienski olefination - Related Concepts (2)

Name reactions

Julia olefination

Julia-Kocienski olefination catalysts

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Exported: Retrieved Related Reaction Results (2,338)	 Reactions	View Results

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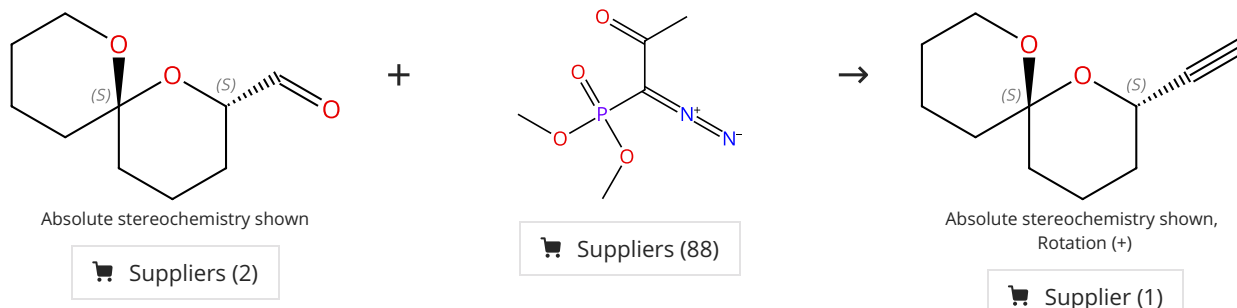
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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



31-339-CAS-7247824

Steps: 1 Yield: 100%

A Concise and Efficient Synthesis of Spiroketal - Application to the Synthesis of SPIKET-P and a Spiroketal from Bactrocera Species

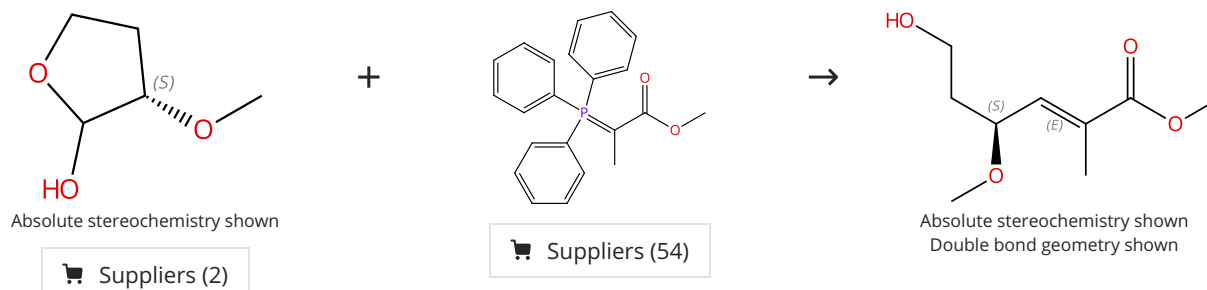
By: Tomas, Loic; et al

European Journal of Organic Chemistry (2013), 2013(5), 915-920.

1.1 **Reagents:** Potassium carbonate
Solvents: Methanol; 2 h, rt

Scheme 2 (1 Reaction)

Steps: 1 Yield: 99%



31-339-CAS-5720616

Steps: 1 Yield: 99%

Total Synthesis of lejimalide B

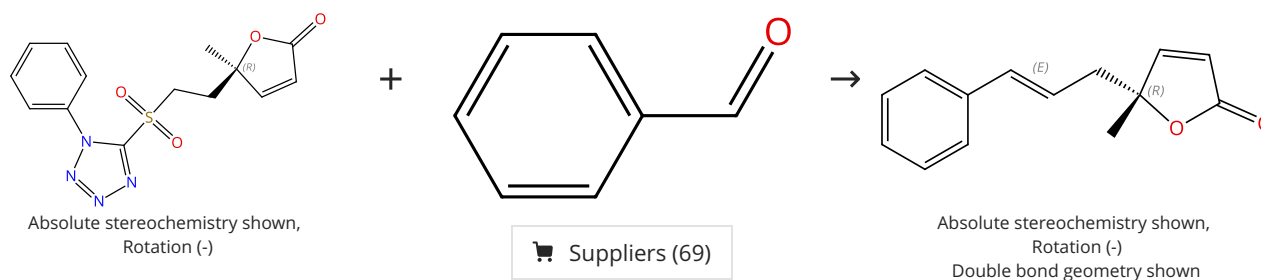
By: Chen, Qingshou; et al

Journal of Organic Chemistry (2011), 76(13), 5157-5169.

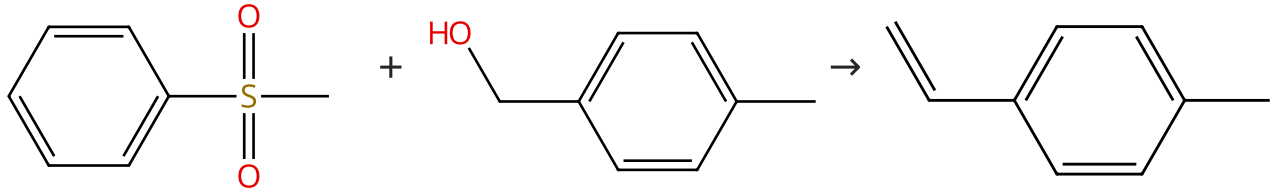
1.1 **Solvents:** Toluene; 3 h, 90 °C
 Experimental Protocols

Scheme 3 (1 Reaction)

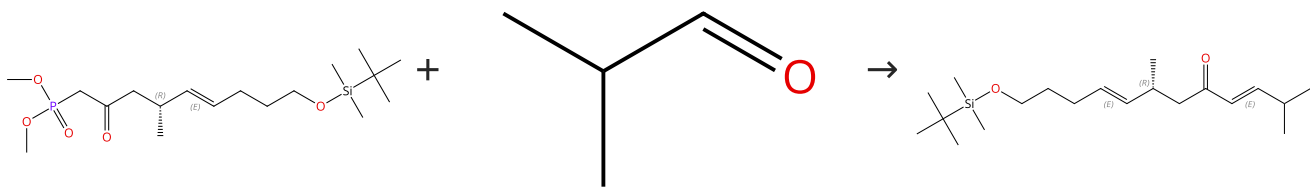
Steps: 1 Yield: 99%



31-339-CAS-15912593	Steps: 1 Yield: 99%	Catalytic asymmetric formal γ-allylation of deconjugated butenolides
1.1 Reagents: Lithium bis(trimethylsilyl)amide Solvents: Tetrahydrofuran; 5 min, -78 °C; 45 min, -78 °C		By: Simlandy, Amit K.; et al
1.2 Solvents: Tetrahydrofuran; 5 min, -78 °C; 40 min, -78 °C; -78 °C → 25 °C; 2 h, 25 °C		Organic & Biomolecular Chemistry (2016), 14(24), 5659-5664.
1.3 Reagents: Ammonium chloride Solvents: Water		
Experimental Protocols		

Scheme 4 (1 Reaction)		Steps: 1 Yield: 99%
		
 Suppliers (77)	 Suppliers (79)	 Suppliers (66)

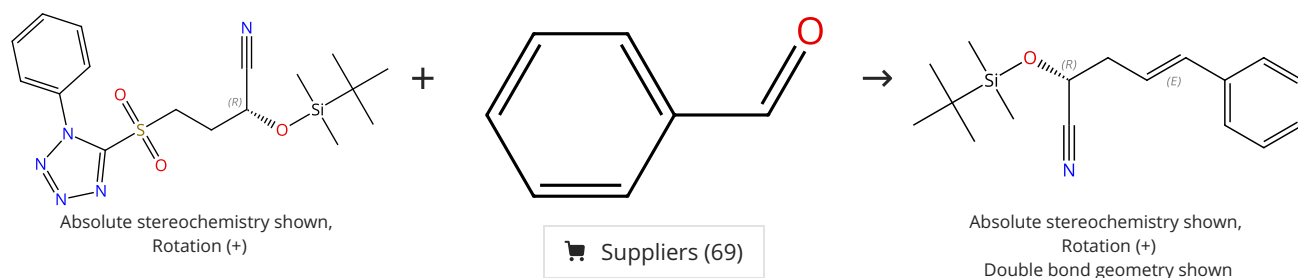
31-339-CAS-18289991	Steps: 1 Yield: 99%	(E)-Specific direct Julia-olefination of aryl alcohols without extra reducing agents promoted by bases
1.1 Reagents: Sodium hydride; 10 h, 135 °C		By: Yao, Chuan-Zhi; et al
Experimental Protocols		Chemical Communications (Cambridge, United Kingdom) (2015), 51(36), 7729-7732.

Scheme 5 (2 Reactions)		Steps: 1 Yield: 99%
		
Absolute stereochemistry shown, Rotation (-) Double bond geometry shown	 Suppliers (74)	Absolute stereochemistry shown, Rotation (-) Double bond geometry shown

31-339-CAS-16194445	Steps: 1 Yield: 99%	Stereodivergent Synthesis and Configurational Assignment of the C1-C15 Segment of Amphirionin-5
1.1 Reagents: Diisopropylethylamine, Lithium chloride Solvents: Acetonitrile; 15 h, rt		By: Kanto, Moemi; et al
1.2 Reagents: Ammonium chloride Solvents: Water		Journal of Organic Chemistry (2016), 81(19), 9105-9121.
Experimental Protocols		
31-339-CAS-5699259	Steps: 1 Yield: 99%	Synthetic Studies on Amphirionin-5: Stereochemical Assignment/Reassignment of the C1-C9 Portion through Stereodivergent Synthesis
1.1 Reagents: Diisopropylethylamine, Lithium chloride Solvents: Acetonitrile; 15 h, rt		By: Kanto, Moemi; et al
1.2 Reagents: Ammonium chloride Solvents: Water; rt		Organic Letters (2016), 18(1), 112-115.
Experimental Protocols		

Scheme 6 (2 Reactions)

Steps: 1 Yield: 99%



31-339-CAS-19364056

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Sodium bis(trimethylsilyl)amide
Solvents: Tetrahydrofuran; 45 min, -78 °C
- 1.2 **Solvents:** Tetrahydrofuran; 3 h, -78 °C
- 1.3 **Reagents:** Ammonium chloride
Solvents: Water; -78 °C; -78 °C → rt

Experimental Protocols

Unified Strategy for 1,5,9- and 1,5,7-Triols via Configuration-Encoded 1,5-Polyol Synthesis: Enantioselective Preparation of γ -Sulfonyl- α -silyloxyaldehydes and Iterative Julia-Kocienski Coupling

By: Friedrich, Ryan M.; et al

Journal of Organic Chemistry (2018), 83(22), 13636-13649.

31-339-CAS-13537374

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Sodium bis(trimethylsilyl)amide
Solvents: Tetrahydrofuran; 45 min, -78 °C
- 1.2 **Solvents:** Tetrahydrofuran; 3 h, -78 °C
- 1.3 **Reagents:** Ammonium chloride
Solvents: Water; -78 °C → rt

Experimental Protocols

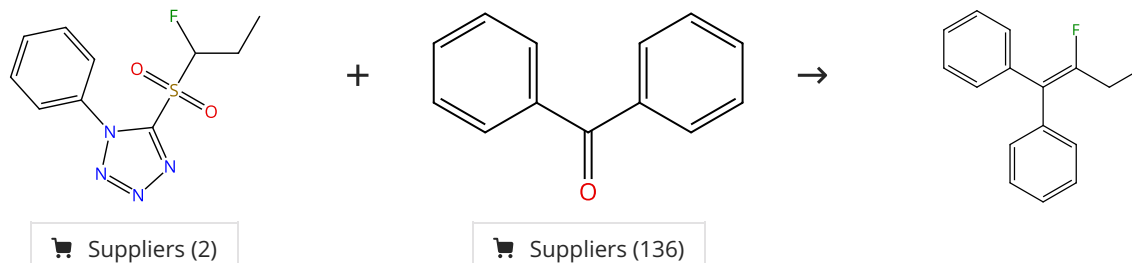
Versatile Configuration-Encoded Strategy for Rapid Synthesis of 1,5-Polyol Stereoisomers

By: Friestad, Gregory K.; et al

Organic Letters (2010), 12(21), 5016-5019.

Scheme 7 (1 Reaction)

Steps: 1 Yield: 99%



31-339-CAS-13436998

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Lithium bis(trimethylsilyl)amide
Solvents: Tetrahydrofuran; 2 h, 0 °C

Experimental Protocols

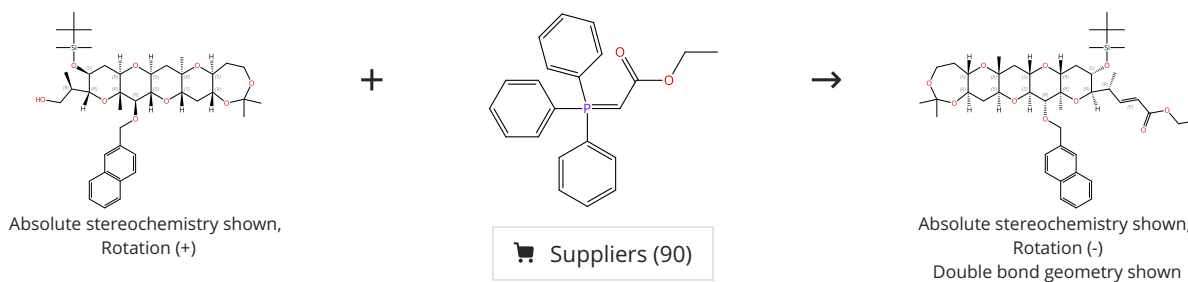
Fluorinated 1-Phenyl-1H-tetrazol-5-yl Sulfone Derivatives as General Reagents for Fluoroalkylidene Synthesis

By: Ghosh, Arun K.; et al

Journal of Organic Chemistry (2009), 74(22), 8531-8540.

Scheme 8 (1 Reaction)

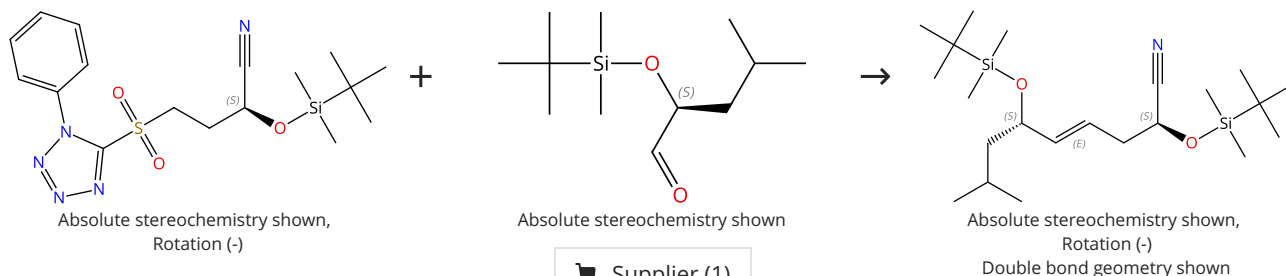
Steps: 1 Yield: 99%



31-339-CAS-13181371	Steps: 1 Yield: 99%	Total Synthesis and Complete Stereostructure of Gambieric Acid A
1.1 Reagents: Triethylamine, Sulfur trioxide-pyridine Solvents: Dimethyl sulfoxide, Dichloromethane; 1 h, 0 °C		By: Fuwa, Haruhiko; et al
1.2 Solvents: Toluene; 20 h, 100 °C		Journal of the American Chemical Society (2012), 134(29), 11984-11987.
Experimental Protocols		

Scheme 9 (2 Reactions)

Steps: 1 Yield: 99%



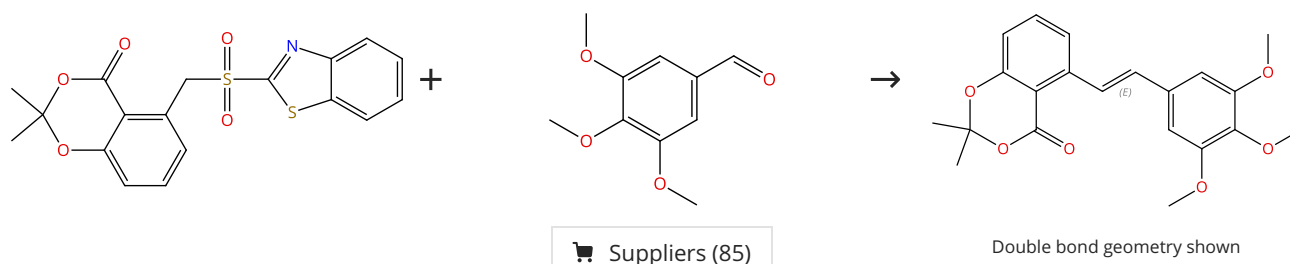
Supplier (1)

31-339-CAS-19364059	Steps: 1 Yield: 99%	Unified Strategy for 1,5,9- and 1,5,7-Triols via Configuration-Encoded 1,5-Polyol Synthesis: Enantioselective Preparation of γ-Sulfonyl-α-silyloxyaldehydes and Iterative Julia-Kocienski Coupling
1.1 Reagents: Potassium bis(trimethylsilyl)amide Solvents: Tetrahydrofuran; 45 min, -78 °C		By: Friedrich, Ryan M.; et al
1.2 Solvents: Tetrahydrofuran; 3 h, -78 °C		Journal of Organic Chemistry (2018), 83(22), 13636-13649.
1.3 Reagents: Ammonium chloride Solvents: Water; -78 °C; -78 °C $\rightarrow$ rt		
Experimental Protocols		

31-339-CAS-4406976	Steps: 1 Yield: 99%	Versatile Configuration-Encoded Strategy for Rapid Synthesis of 1,5-Polyol Stereoisomers
1.1 Reagents: Sodium bis(trimethylsilyl)amide Solvents: Tetrahydrofuran; 45 min, -78 °C		By: Friestad, Gregory K.; et al
1.2 Solvents: Tetrahydrofuran; 3 h, -78 °C		Organic Letters (2010), 12(21), 5016-5019.
1.3 Reagents: Ammonium chloride Solvents: Water; -78 °C $\rightarrow$ rt		
Experimental Protocols		

Scheme 10 (1 Reaction)

Steps: 1 Yield: 99%

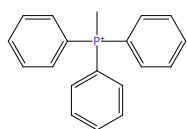


Suppliers (85)

31-333-CAS-16619252	Steps: 1 Yield: 99%	Valuable building block for the synthesis of lunularic acid, hydrangeic acid and their analogues
1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 5 min, 0 °C		By: Mukkamala, Ramesh; et al
1.2 0 °C $\rightarrow$ rt; 4 - 5 h, rt		Natural Product Research (2017), 31(9), 1085-1090.
1.3 Reagents: Ammonium chloride Solvents: Water		
Experimental Protocols		

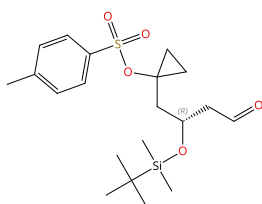
Scheme 11 (1 Reaction)

Steps: 1 Yield: 98%

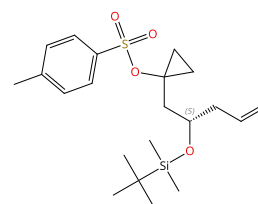
• Br⁻

Suppliers (83)

+

Absolute stereochemistry shown,
Rotation (+)

→

Absolute stereochemistry shown,
Rotation (+)

31-339-CAS-4514157

Steps: 1 Yield: 98%

**Enantioselective total synthesis of the proposed structure of
macrolide iriomoteolide-1b**

By: Ye, Zhengqing; et al

Tetrahedron (2011), 67(33), 5979-5989.

1.1 Reagents: Sodium bis(trimethylsilyl)amide

Solvents: Tetrahydrofuran; 30 min, 0 °C

1.2 Solvents: Tetrahydrofuran; 30 min, 0 °C

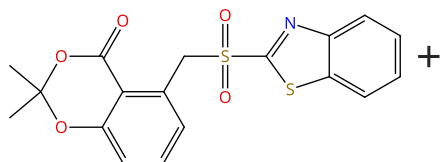
1.3 Reagents: Ammonium chloride

Solvents: Water; 0 °C

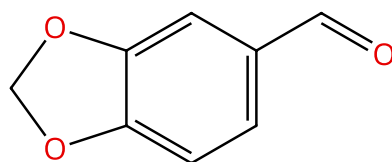
Experimental Protocols

Scheme 12 (1 Reaction)

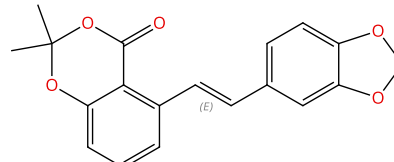
Steps: 1 Yield: 98%



+



→



Suppliers (42)

Double bond geometry shown

31-333-CAS-16619245

Steps: 1 Yield: 98%

**Valuable building block for the synthesis of lunularic acid,
hydrangeic acid and their analogues**

By: Mukkamala, Ramesh; et al

Natural Product Research (2017), 31(9), 1085-1090.

1.1 Reagents: Sodium hydride

Solvents: Dimethylformamide; 5 min, 0 °C

1.2 0 °C → rt; 4 - 5 h, rt

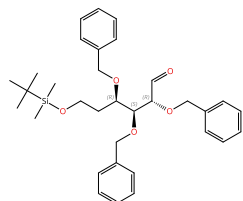
1.3 Reagents: Ammonium chloride

Solvents: Water

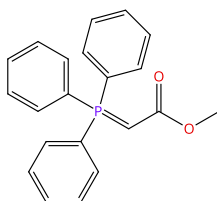
Experimental Protocols

Scheme 13 (1 Reaction)

Steps: 1 Yield: 98%

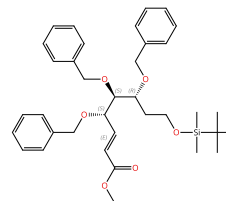
Absolute stereochemistry shown,
Rotation (+)

+



Suppliers (86)

→

Absolute stereochemistry shown,
Rotation (+)
Double bond geometry shown

31-339-CAS-335227

Steps: 1 Yield: 98%

**Stereoselective Synthesis and Absolute Configuration of the
C1'-C25' Fragment of Symbiodinide**

By: Takamura, Hiroyoshi; et al

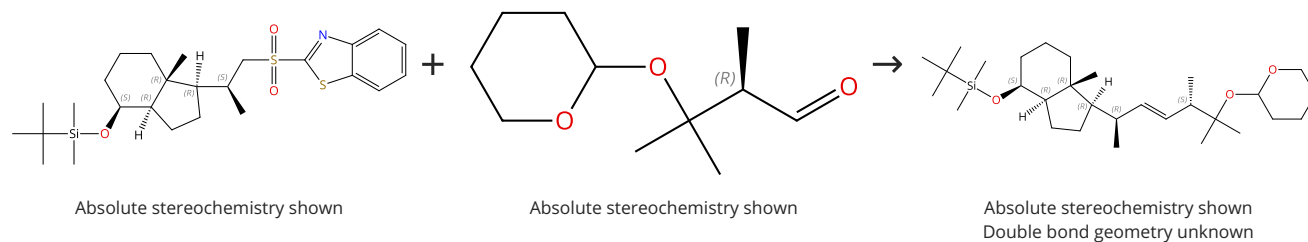
Journal of Organic Chemistry (2009), 74(17), 6658-6666.

1.1 Solvents: Benzene; 2 h, 40 °C

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 98%



31-339-CAS-773874

Steps: 1 Yield: 98%

Process for the preparation of the "Windaus 25-hydroxy ketone" from Vitamin D₂ via Julia-Kocienski reaction

Assignee: Universidad de Vigo

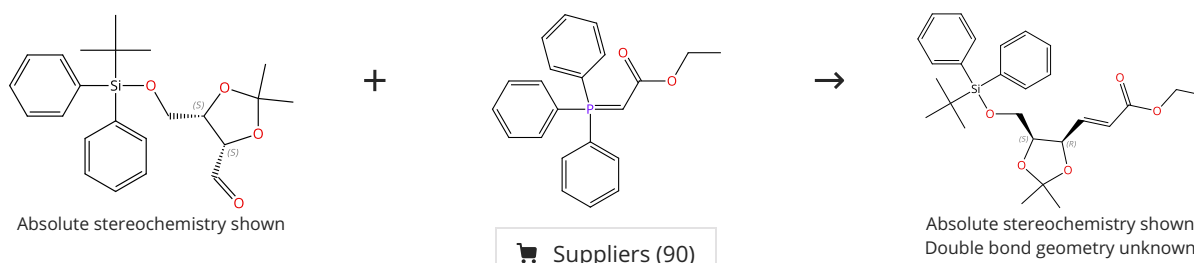
Spain, ES2380477 A1 2012-05-14

[PatentPak](#) available

- 1.1 **Solvents:** Tetrahydrofuran; rt → -78 °C
 1.2 **Reagents:** Lithium bis(trimethylsilyl)amide
Solvents: Tetrahydrofuran; 1 h, -78 °C
 1.3 **Solvents:** Tetrahydrofuran; -78 °C; -78 °C → rt; overnight, rt
 1.4 **Reagents:** Ammonium chloride
Solvents: Water; rt

Scheme 15 (1 Reaction)

Steps: 1 Yield: 98%



31-339-CAS-9711159

Steps: 1 Yield: 98%

Asymmetric Total Synthesis of Bioactive Natural Lipid Mycalol

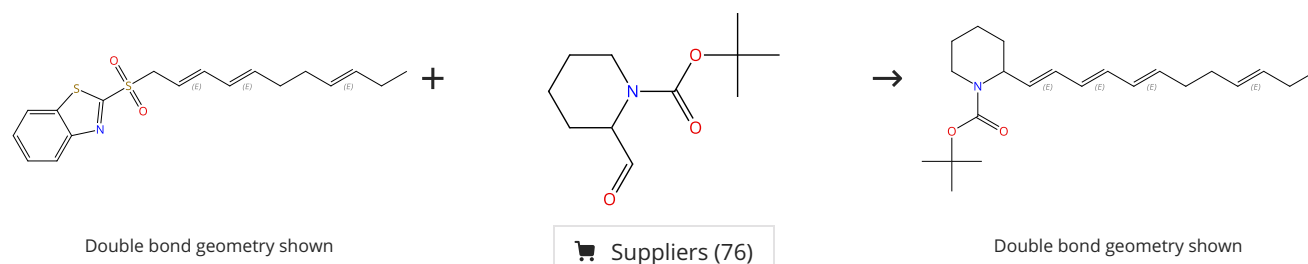
By: Das, Subhendu; et al

Journal of Organic Chemistry (2015), 80(12), 6467-6489.

- 1.1 **Solvents:** Dichloromethane; 0 °C; 0 °C → rt; 12 h, rt
 Experimental Protocols

Scheme 16 (1 Reaction)

Steps: 1 Yield: 98%



31-339-CAS-5028035

Steps: 1 Yield: 98%

Synthesis of a model compound of coryden dramine A via a Julia coupling

By: McCrea-Hendrick, Maddy; et al

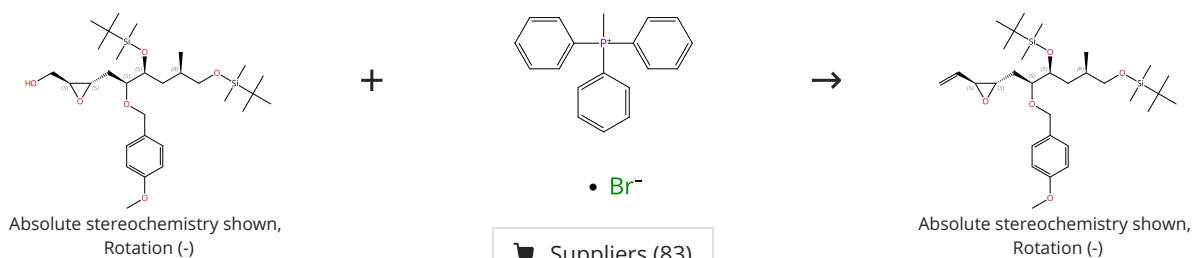
Synthetic Communications (2009), 39(20), 3611-3620.

- 1.1 **Reagents:** 18-Crown-6, Potassium bis(trimethylsilyl)amide
Solvents: Tetrahydrofuran; 1 h, -80 °C
 1.2 **Solvents:** Tetrahydrofuran; -80 °C; 2.5 h, -80 °C; overnight, -80 °C → rt
 1.3 **Reagents:** Ammonium chloride
Solvents: Water

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (83)

31-339-CAS-18213283

Steps: 1 Yield: 98%

A stereoselective total synthesis of 7,8-O-isopropylidene iriomoteolide-3a

By: Zhang, Yao; et al

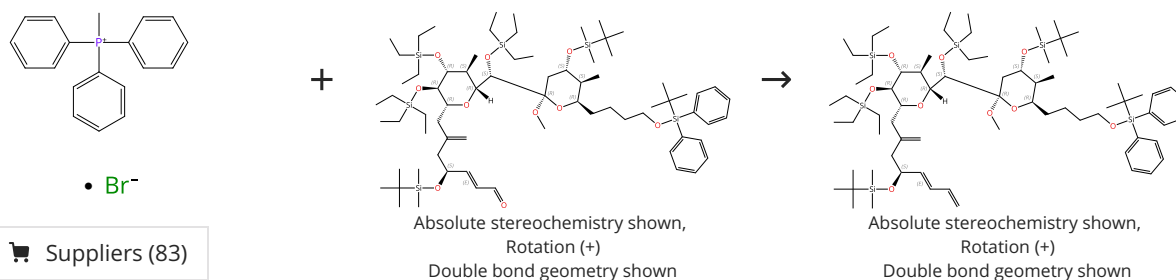
Organic & Biomolecular Chemistry (2011), 9(12), 4518-4526.

1.1 **Reagents:** Dimethyl sulfoxide, Diisopropylethylamine, Sulfur trioxide-pyridine complex**Solvents:** Dichloromethane; 1 h, 0 °C1.2 **Reagents:** Sodium bis(trimethylsilyl)amide**Solvents:** Tetrahydrofuran; 30 min, 0 °C1.3 **Solvents:** Tetrahydrofuran; -40 °C; 3 h, -40 °C

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (83)

31-339-CAS-4459538

Steps: 1 Yield: 98%

Spongipyran synthetic studies. Total synthesis of (+)-spongi statin 2

By: Smith, Amos B. III; et al

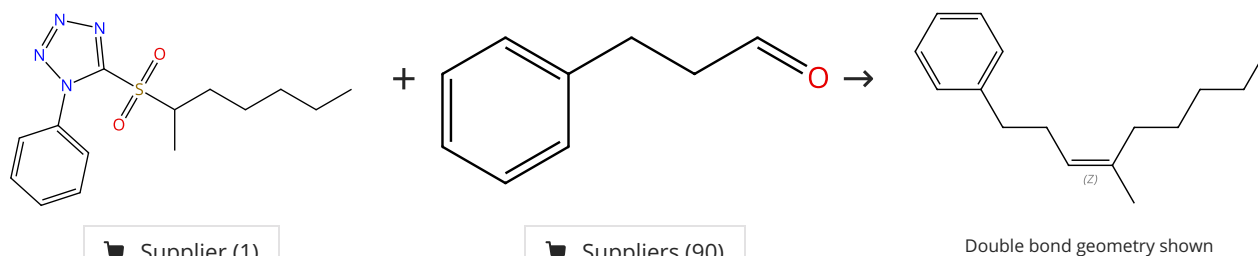
Tetrahedron (2009), 65(33), 6470-6488.

1.1 **Reagents:** Butyllithium**Solvents:** Tetrahydrofuran, Hexane; 0 °C; 0.5 h, 0 °C; 0 °C → -78 °C1.2 **Solvents:** Tetrahydrofuran; 20 min, -78 °C; -78 °C → 0 °C; 10 min, 0 °C1.3 **Reagents:** Ammonium chloride

Experimental Protocols

Scheme 19 (1 Reaction)

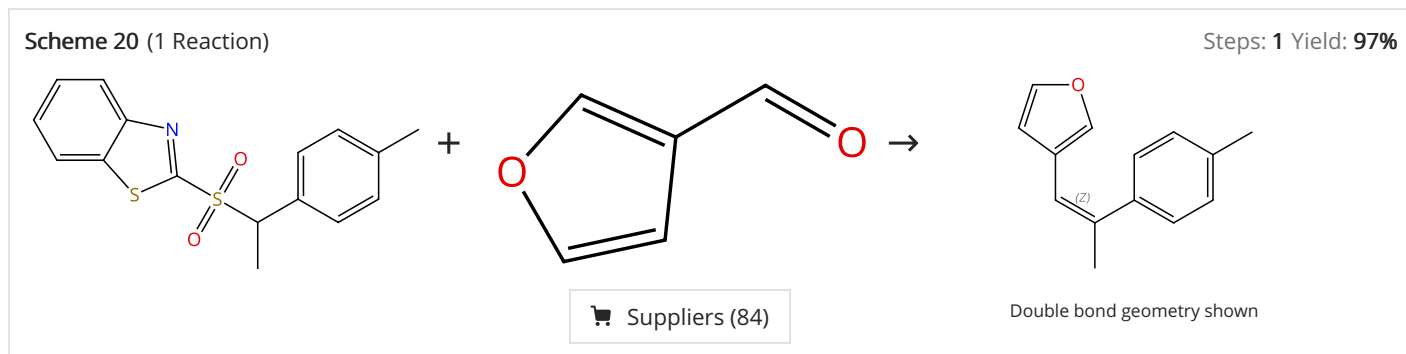
Steps: 1 Yield: 97%



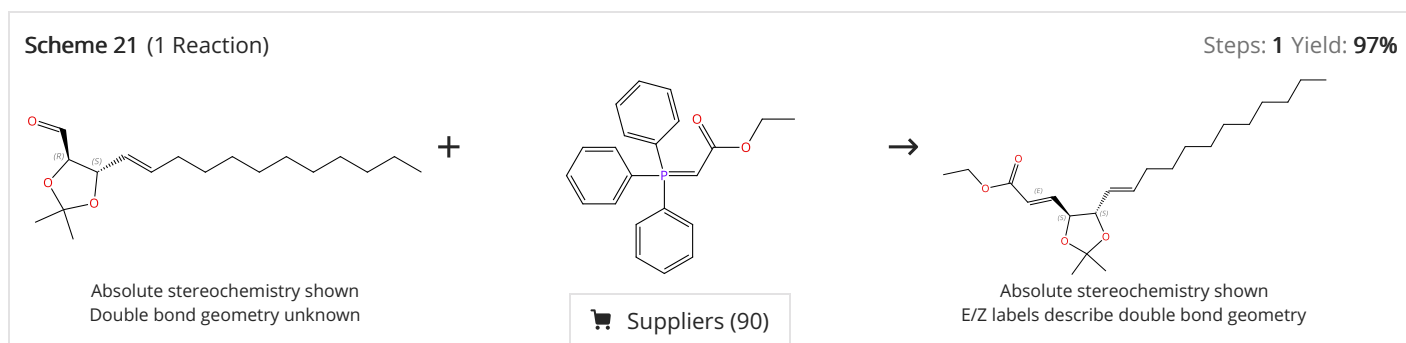
Supplier (1)

Suppliers (90)

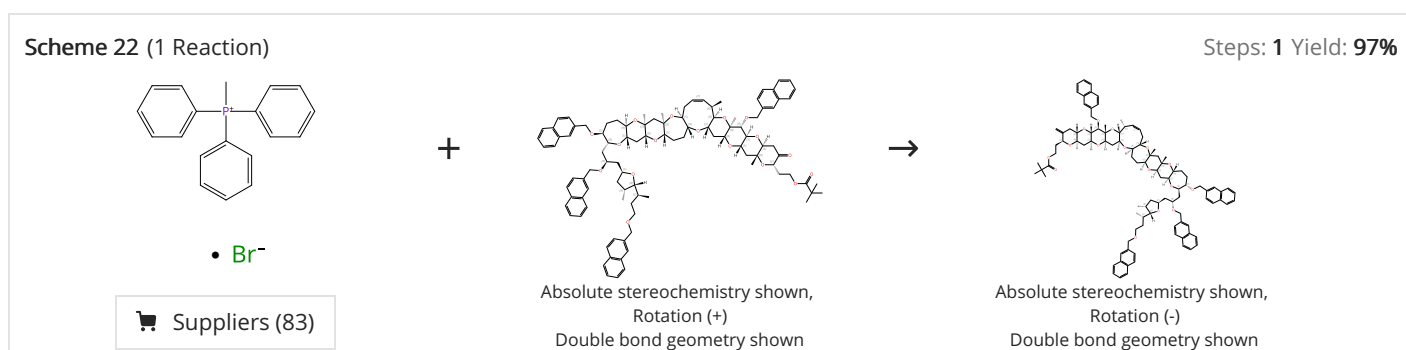
31-339-CAS-9699807	Steps: 1 Yield: 97%	Simple protocol for enhanced (E)-selectivity in Julia-Kocienski reaction
1.1 Reagents: 12-Crown-4, Lithium bis(trimethylsilyl)amide Solvents: Tetrahydrofuran; 10 s, -78 °C; 30 s, -78 °C		By: Pospisil, Jiri
1.2 Solvents: Tetrahydrofuran; 30 min, -78 °C		Tetrahedron Letters (2011), 52(18), 2348-2352.
1.3 Reagents: Ammonium chloride Solvents: Water; -78 °C		
Experimental Protocols		



31-333-CAS-8620857	Steps: 1 Yield: 97%	Markovnikov-Selective Hydrothiolation of Styrenes: Application to the Synthesis of Stereodefined Trisubstituted Olefins
1.1 Reagents: Lithium diisopropylamide Solvents: Tetrahydrofuran; -78 °C; 15 - 18 h, -78 °C → rt		By: Savolainen, Markku A.; et al
1.2 Reagents: Ammonium chloride Solvents: Water		Organic Letters (2013), 15(14), 3802-3804.
Experimental Protocols		



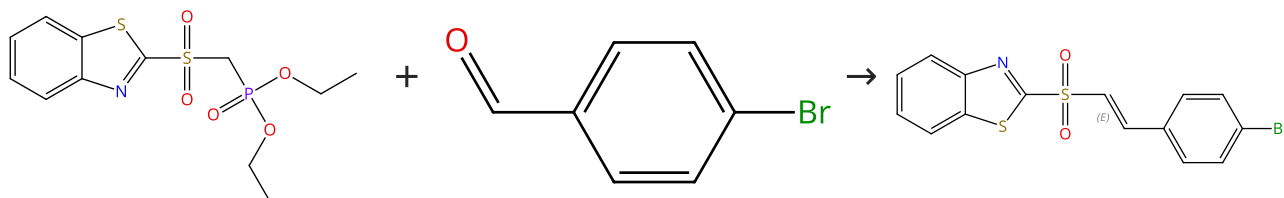
31-339-CAS-15267687	Steps: 1 Yield: 97%	An Efficient Stereoselective Approach for the Synthesis of (+)-(4S,5S)-Muricatacin
1.1 Solvents: Benzene; rt; 2 h, rt		By: Srinivas, Chiguru; et al
Experimental Protocols		Helvetica Chimica Acta (2011), 94(4), 669-674.



31-339-CAS-19443601	Steps: 1 Yield: 97%	Total Synthesis and Complete Stereostructure of Gambieric Acid A
1.1 Reagents: Sodium bis(trimethylsilyl)amide Solvents: Tetrahydrofuran; 30 min, 0 °C		By: Fuwa, Haruhiko; et al
1.2 Solvents: Tetrahydrofuran; 3 h, -78 °C → 0 °C		Journal of the American Chemical Society (2012), 134(29), 11984-11987.
Experimental Protocols		

Scheme 23 (1 Reaction)

Steps: 1 Yield: 97%



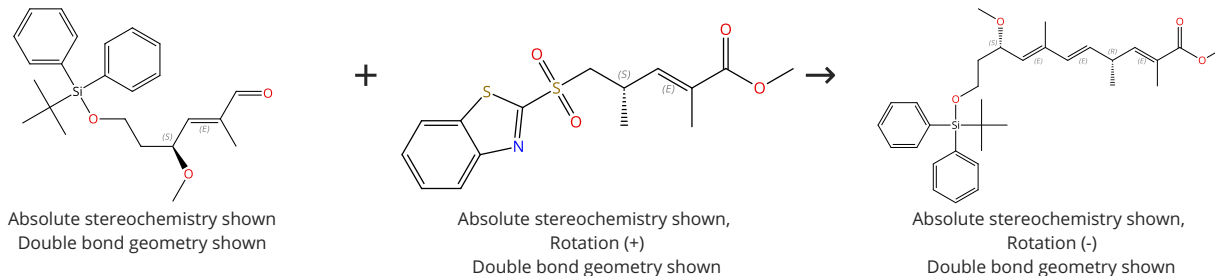
Suppliers (91)

Double bond geometry shown

31-614-CAS-42153592	Steps: 1 Yield: 97%	Asymmetric Synthesis of Chiral 1, 3-Disubstituted Allylsilanes via Copper(I)-Catalyzed 1, 4-Conjugate Silylation of α, β-Unsaturated Sulfones and Subsequent Julia-Kocienski Olefination
1.1 Reagents: Potassium carbonate Solvents: Dichloromethane; 4 - 10 h, rt		By: Wang, Xian-Liang; et al
Experimental Protocols		Chinese Journal of Chemistry (2021), 39(7), 1916-1922.

Scheme 24 (1 Reaction)

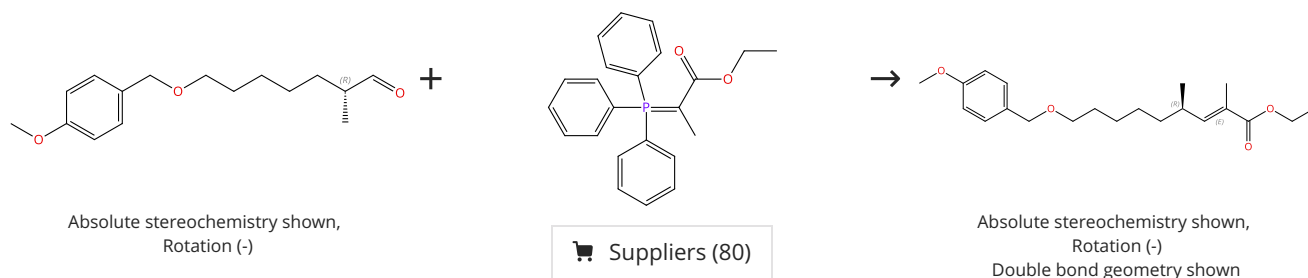
Steps: 1 Yield: 97%



31-339-CAS-5992161	Steps: 1 Yield: 97%	Total Synthesis of lejimalide B
1.1 Reagents: Lithium bis(trimethylsilyl)amide Solvents: Tetrahydrofuran; -78 °C; 30 min, -78 °C; 1.5 h, -78 °C → 23 °C; 30 min, 23 °C; 23 °C → -78 °C		By: Chen, Qingshou; et al
1.2 Reagents: Ammonium chloride Solvents: Water; -78 °C; 30 min, -78 °C → 23 °C		Journal of Organic Chemistry (2011), 76(13), 5157-5169.
Experimental Protocols		

Scheme 25 (1 Reaction)

Steps: 1 Yield: 97%



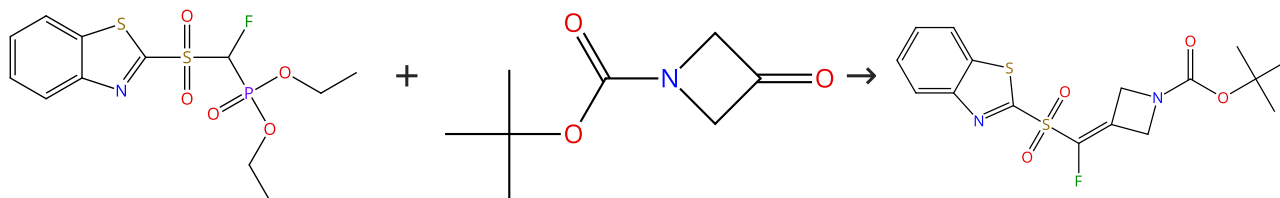
Suppliers (80)

 Absolute stereochemistry shown,
Rotation (-)
Double bond geometry shown

31-339-CAS-14856084	Steps: 1 Yield: 97%	Formal Synthesis of (+)-Sorangicin A
1.1 Solvents: Benzene; overnight, reflux		By: Crimmins, Michael T.; et al
Experimental Protocols		Organic Letters (2011), 13(17), 4712-4715.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 96%



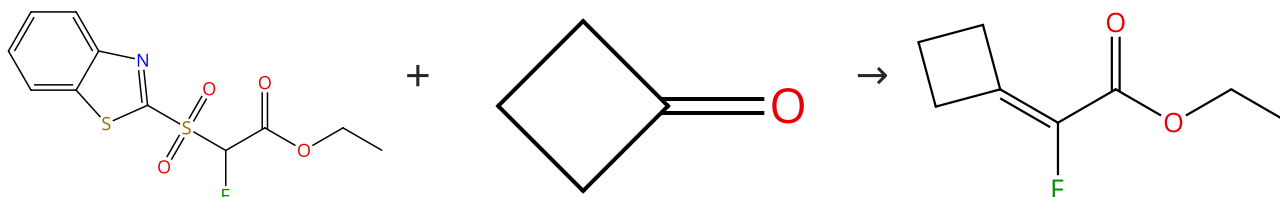
Suppliers (99)

Supplier (1)

31-333-CAS-7440632	Steps: 1 Yield: 96%	Synthesis of Fluorine-Containing 3,3-Disubstituted Oxetanes and Alkylidene Oxetanes
1.1 Reagents: Triethylamine, Lithium chloride Solvents: Tetrahydrofuran; 0 °C; overnight, 20 °C		By: Laporte, Romain; et al
1.2 Reagents: Ammonium chloride Solvents: Water		European Journal of Organic Chemistry (2015), 2015(14), 3121-3128.
Experimental Protocols		

Scheme 27 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (4)

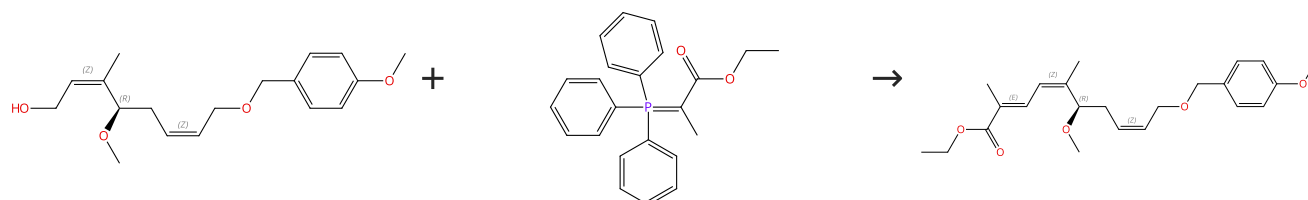
Suppliers (81)

Suppliers (3)

31-331-CAS-13659046	Steps: 1 Yield: 96%	Synthesis of Fluorine-Containing 3,3-Disubstituted Oxetanes and Alkylidene Oxetanes
1.1 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene Solvents: Tetrahydrofuran; 0 °C; overnight, 20 °C		By: Laporte, Romain; et al
1.2 Reagents: Ammonium chloride Solvents: Water		European Journal of Organic Chemistry (2015), 2015(14), 3121-3128.
Experimental Protocols		

Scheme 28 (1 Reaction)

Steps: 1 Yield: 96%



Absolute stereochemistry shown,
Rotation (+)
Double bond geometry shown

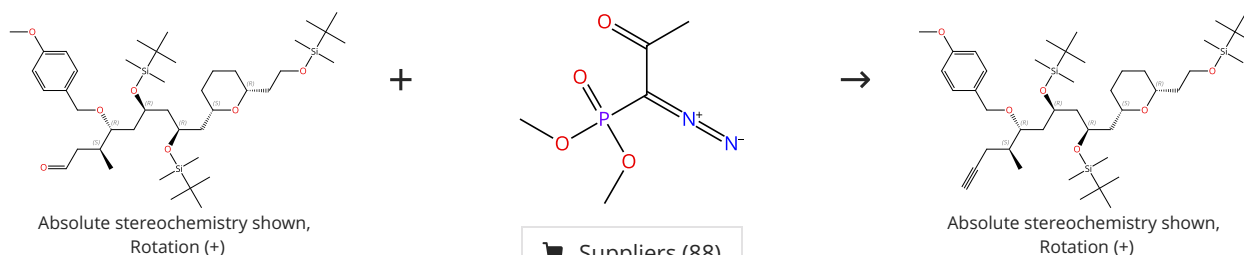
Suppliers (80)

Absolute stereochemistry shown,
Rotation (+)
Double bond geometry shown

31-339-CAS-21301047 Steps: 1 Yield: 96% 1.1 Reagents: Manganese oxide (MnO ₂) Solvents: Benzene; 1.5 h, 50 °C 1.2 9 h, 50 °C Experimental Protocols	Efficient Total Synthesis of Bongkreic Acid and Apoptosis Inhibitory Activity of Its Analogues By: Matsumoto, Kenji; et al Chemistry - A European Journal (2015), 21(32), 11590-11602.
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Scheme 29 (1 Reaction)

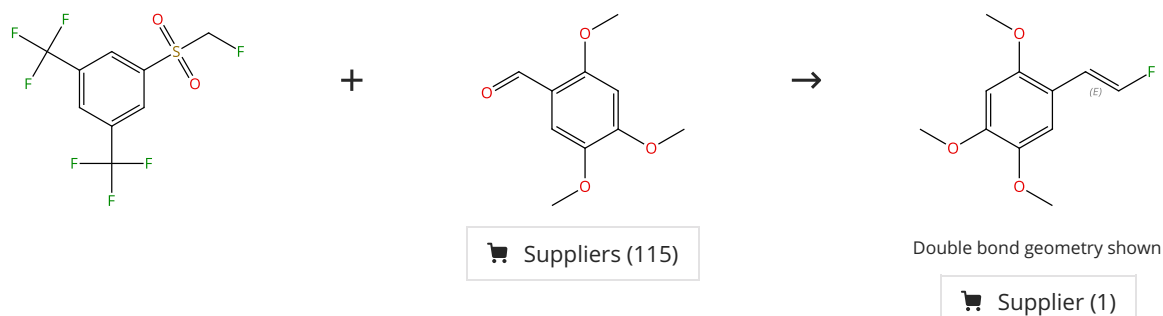
Steps: 1 Yield: 96%



31-339-CAS-9293277 Steps: 1 Yield: 96% 1.1 Reagents: Potassium carbonate Solvents: Methanol; 14 h, rt Experimental Protocols	Second-Generation Total Synthesis of Spirastrellolide F Methyl Ester: The Alkyne Route By: Benson, Stefan; et al Angewandte Chemie, International Edition (2011), 50(37), 8739-8744, S8739/1-S8739/50.
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Scheme 30 (1 Reaction)

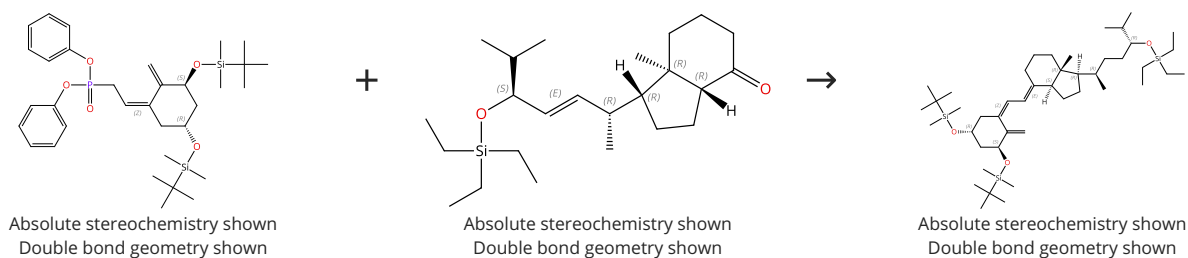
Steps: 1 Yield: 96%



31-339-CAS-393371 Steps: 1 Yield: 96% 1.1 Reagents: Potassium hydroxide Solvents: Dimethyl sulfoxide; 2 h, rt	Synthesis of monofluoroalkenes via Julia-Kocienski reaction By: Prakash, G. K. Surya; et al Journal of Fluorine Chemistry (2010), 131(11), 1192-1197.
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Scheme 31 (1 Reaction)

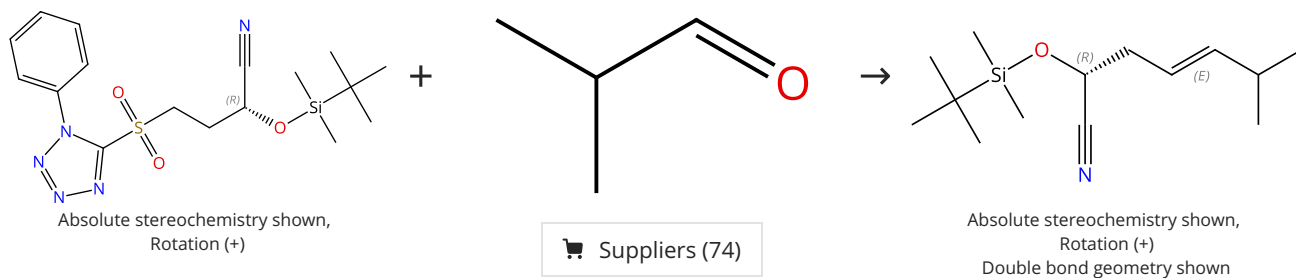
Steps: 1 Yield: 96%



31-242-CAS-23544285	Steps: 1 Yield: 96%	Process Development of Tacalcitol
1.1 Reagents: Butyllithium Solvents: Tetrahydrofuran; 1 h, -65 °C	Experimental Protocols	By: Lee, Seung Jong; et al Organic Process Research & Development (2021), 25(4), 982-987.
1.2 Solvents: Tetrahydrofuran; 1 h, -65 °C; -65 °C → -10 °C; 30 min, -10 °C		
1.3 Reagents: Water		

Scheme 32 (1 Reaction)

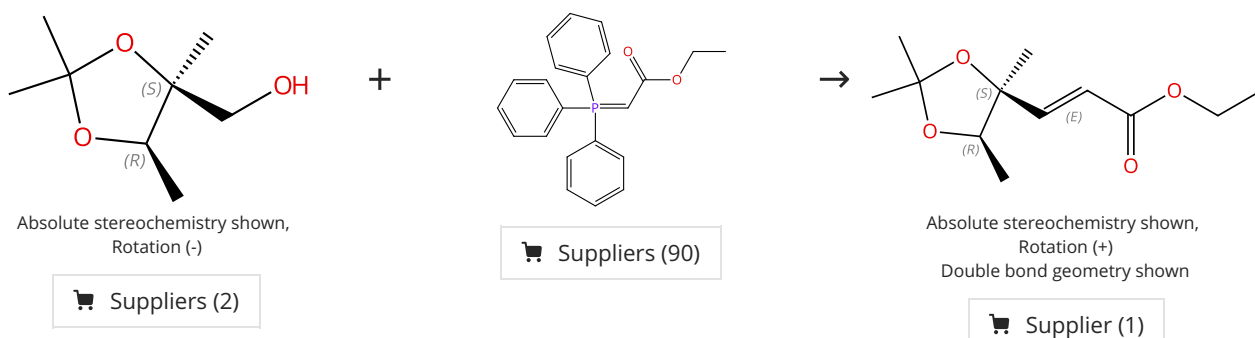
Steps: 1 Yield: 96%



31-339-CAS-125079	Steps: 1 Yield: 96%	Versatile Configuration-Encoded Strategy for Rapid Synthesis of 1,5-Polyol Stereoisomers
1.1 Reagents: Sodium bis(trimethylsilyl)amide Solvents: Tetrahydrofuran; 45 min, -78 °C	Experimental Protocols	By: Friestad, Gregory K.; et al Organic Letters (2010), 12(21), 5016-5019.
1.2 Solvents: Tetrahydrofuran; 3 h, -78 °C		
1.3 Reagents: Ammonium chloride Solvents: Water; -78 °C → rt		

Scheme 33 (1 Reaction)

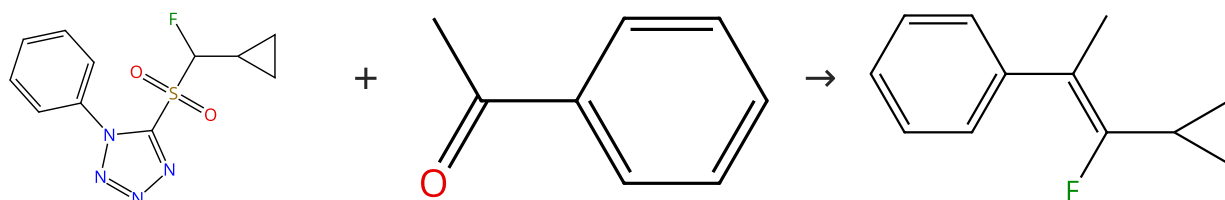
Steps: 1 Yield: 96%



31-339-CAS-18294680	Steps: 1 Yield: 96%	Studies directed towards the total synthesis of polyether antibiotic ionomycin: asymmetric synthesis of fragments C (24)-C(32) and C(1)-C(23)
1.1 Reagents: Dimethyl sulfoxide, Oxalyl chloride Solvents: Dichloromethane; 20 min, -78 °C; 30 min, -78 °C	Experimental Protocols	By: Yadav, J. S.; et al Tetrahedron (2015), 71(40), 7539-7549.
1.2 Solvents: Dichloromethane; -78 °C; 30 min, -78 °C		
1.3 Reagents: Triethylamine; -78 °C; -78 °C → rt; 30 min, rt		
1.4 Solvents: Dichloromethane; 12 h, rt		
1.5 Reagents: Water; rt		

Scheme 34 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (105)

31-339-CAS-4398596

Steps: 1 Yield: 96%

Fluorinated 1-Phenyl-1H-tetrazol-5-yl Sulfone Derivatives as General Reagents for Fluoroalkylidene Synthesis

By: Ghosh, Arun K.; et al

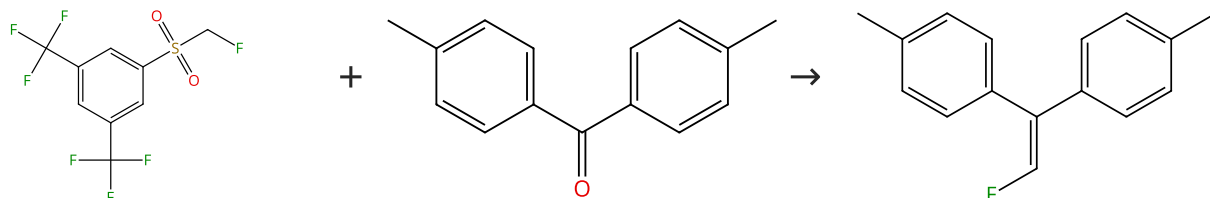
Journal of Organic Chemistry (2009), 74(22), 8531-8540.

1.1 **Reagents:** Lithium bis(trimethylsilyl)amide
Solvents: Tetrahydrofuran; 3 h, 0 °C

Experimental Protocols

Scheme 35 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (72)

Supplier (1)

31-339-CAS-9035202

Steps: 1 Yield: 96%

Synthesis of monofluoroalkenes via Julia-Kocienski reaction

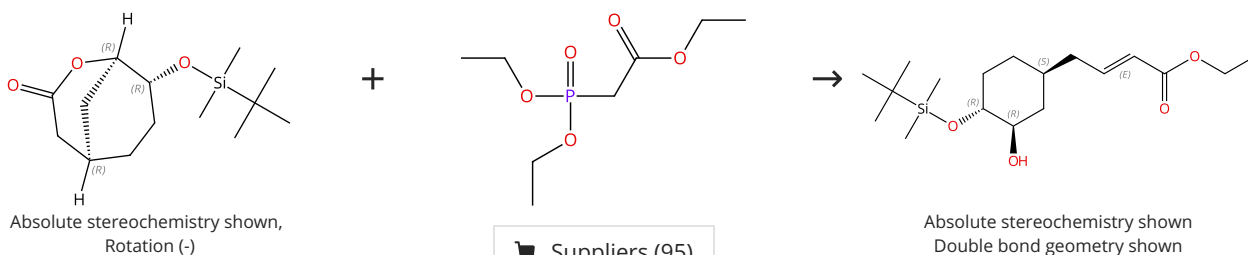
By: Prakash, G. K. Surya; et al

Journal of Fluorine Chemistry (2010), 131(11), 1192-1197.

1.1 **Reagents:** Potassium hydroxide
Solvents: Dimethyl sulfoxide; 2 h, rt

Scheme 36 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (95)

31-339-CAS-9811661

Steps: 1 Yield: 96%

Stereoselective synthesis of C17-C34 fragment of antasc omicin A

By: Vakiti, Jithender Reddy; et al

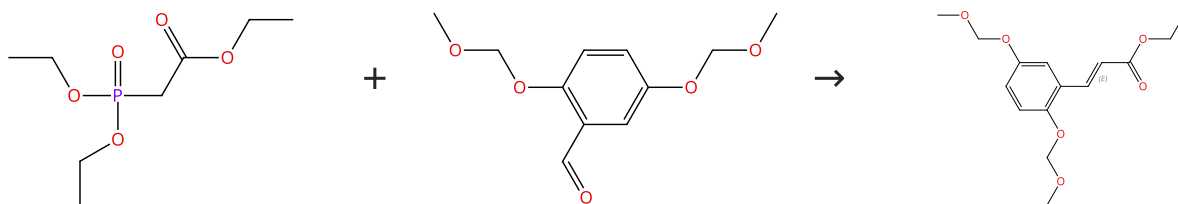
Tetrahedron Letters (2014), 55(47), 6438-6440.

1.1 **Reagents:** Diisobutylaluminum hydride
Solvents: Toluene; 30 min, -78 °C

1.2 **Reagents:** Sodium hydride
Solvents: Tetrahydrofuran; 1 h, -78 °C

Scheme 37 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (95)

Suppliers (7)

Double bond geometry shown

31-339-CAS-12079106

Steps: 1 Yield: 96%

Concise Syntheses and Biological Activities of Ganomycin I and Fornicin A

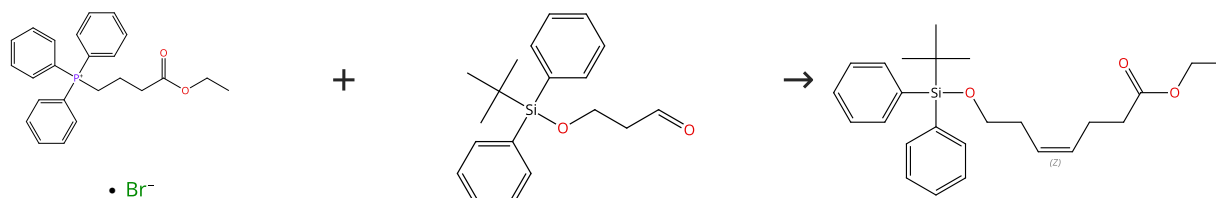
By: Yajima, Arata; et al

European Journal of Organic Chemistry (2014), 2014(4), 731-738.

- 1.1 **Reagents:** Sodium hydride
Solvents: Tetrahydrofuran; 30 min, 0 °C
- 1.2 **Solvents:** Tetrahydrofuran; 1 h, 0 °C

Scheme 38 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (65)

Suppliers (18)

Double bond geometry shown

31-339-CAS-13361532

Steps: 1 Yield: 96%

First total synthesis of labeled E PA and DHA-derived A-type cyclopentenone isoprostanoids: [D₂]-15-A_{3t}-IsoP and [D₂]-17-A_{4t}-NeuroP

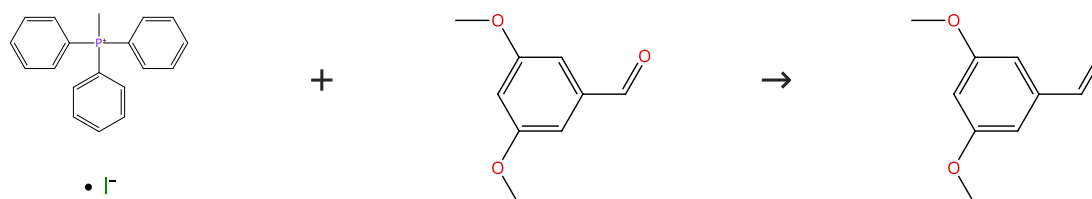
By: Porta, Alessio; et al

Tetrahedron (2014), 70(7), 1484-1491.

- 1.1 **Reagents:** Potassium bis(trimethylsilyl)amide
Solvents: Toluene; rt; 2 h, rt
- 1.2 **Solvents:** Toluene; -100 °C; 30 min, -100 °C
- 1.3 **Reagents:** Ammonium chloride
Solvents: Water

Scheme 39 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (79)

Suppliers (93)

Suppliers (28)

31-339-CAS-9030906

Steps: 1 Yield: 96%

Asymmetric Total Syntheses of Cochliomycin A and Zeaenol

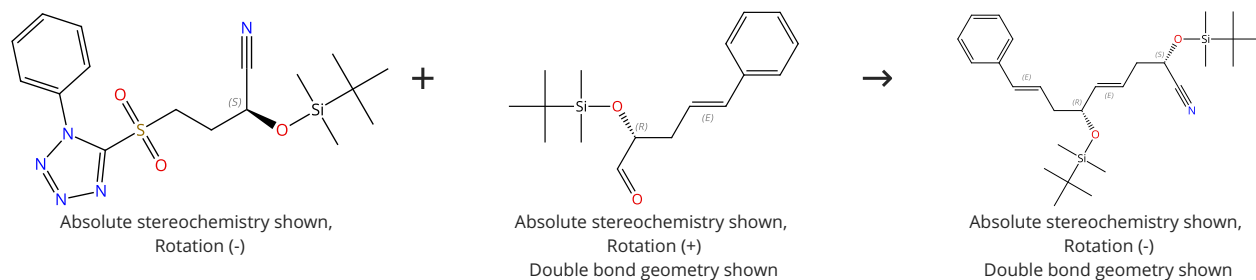
By: Jana, Nandan; et al

European Journal of Organic Chemistry (2012), 2012(23), 4313-4320, S4313/1-S4313/56.

- 1.1 **Reagents:** Potassium *tert*-butoxide
Solvents: Diethyl ether; 2 h, rt; rt → 0 °C
- 1.2 **Solvents:** Diethyl ether; 0 °C; 0 °C → rt
- 1.3 **Solvents:** Methanol; rt

Experimental Protocols

Scheme 40 (2 Reactions)



Steps: 1 Yield: 96%

31-339-CAS-19364062

Steps: 1 Yield: 96%

Unified Strategy for 1,5,9- and 1,5,7-Triols via Configuration-Encoded 1,5-Polyol Synthesis: Enantioselective Preparation of γ -Sulfonyl- α -silyloxyaldehydes and Iterative Julia-Kocienski Coupling

By: Friedrich, Ryan M.; et al

Journal of Organic Chemistry (2018), 83(22), 13636-13649.

1.1 **Reagents:** Potassium bis(trimethylsilyl)amide
Solvents: Tetrahydrofuran; 45 min, -78 °C

1.2 **Solvents:** Tetrahydrofuran; 3 h, -78 °C

1.3 **Reagents:** Ammonium chloride
Solvents: Water; -78 °C; -78 °C $\rightarrow$ rt

Experimental Protocols

31-339-CAS-8671273

Steps: 1 Yield: 96%

Versatile Configuration-Encoded Strategy for Rapid Synthesis of 1,5-Polyol Stereoisomers

By: Friestad, Gregory K.; et al

Organic Letters (2010), 12(21), 5016-5019.

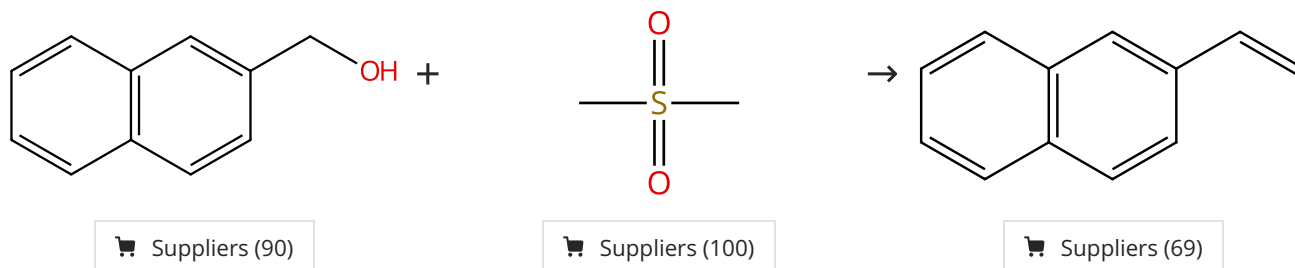
1.1 **Reagents:** Sodium bis(trimethylsilyl)amide
Solvents: Tetrahydrofuran; 45 min, -78 °C

1.2 **Solvents:** Tetrahydrofuran; 3 h, -78 °C

1.3 **Reagents:** Ammonium chloride
Solvents: Water; -78 °C $\rightarrow$ rt

Experimental Protocols

Scheme 41 (1 Reaction)



Steps: 1 Yield: 96%

31-339-CAS-21111104

Steps: 1 Yield: 96%

Direct Olefination of Alcohols with Sulfones by Using Heterogeneous Platinum Catalysts

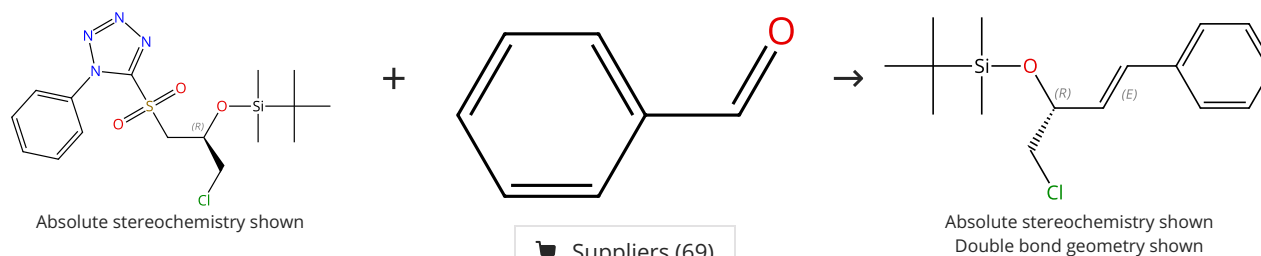
By: Siddiki, S. M. A. Hakim; et al

Chemistry - A European Journal (2016), 22(17), 6111-6119.

1.1 **Reagents:** Potassium *tert*-butoxide
Catalysts: Platinum
Solvents: Toluene; 15 h, reflux

Experimental Protocols

Scheme 42 (1 Reaction)

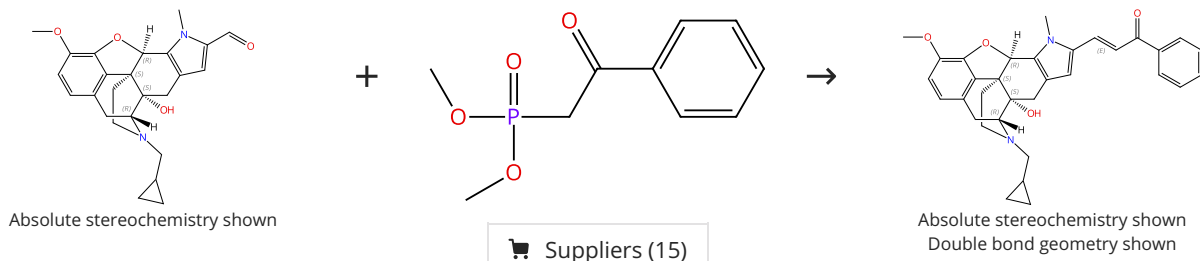


Steps: 1 Yield: 95%

31-339-CAS-17714910 Steps: 1 Yield: 95% 1.1 Reagents: Sodium bis(trimethylsilyl)amide Solvents: Tetrahydrofuran; 0.5 h, -78 °C; 1 h, rt	Synthesis method for chiral secondary allyl alcohol with hydroxy ortho position substituted with halogen atom Assignee: Jiangnan University China, CN106588565 A 2017-04-26 PatentPak available
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Scheme 43 (1 Reaction)

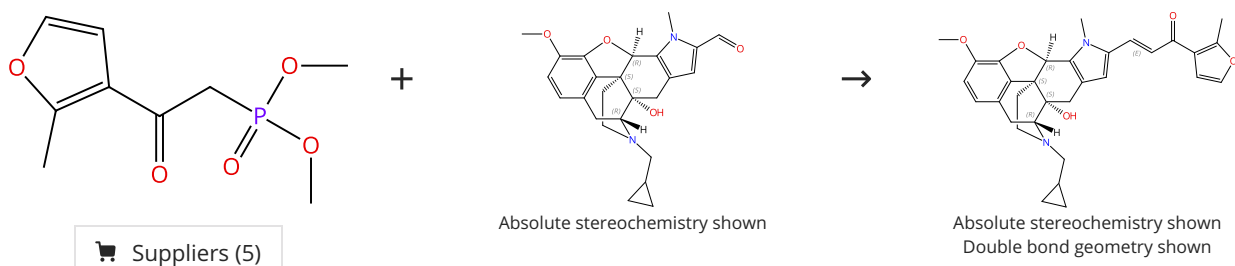
Steps: 1 Yield: 95%



31-339-CAS-11007699 Steps: 1 Yield: 95% 1.1 Reagents: Sodium hydride Solvents: Tetrahydrofuran; 50 °C	Synthesis of pyrrolomorphinan derivatives as κ opioid agonists By: Fujii, Hideaki; et al Bioorganic & Medicinal Chemistry Letters (2010), 20(17), 5035-5038.
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Scheme 44 (1 Reaction)

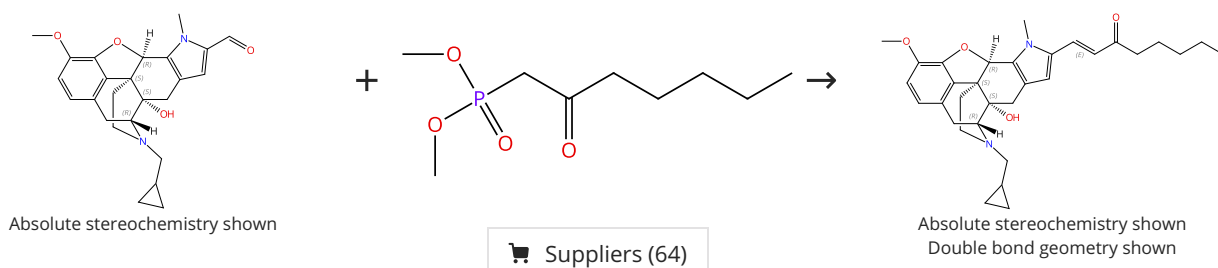
Steps: 1 Yield: 95%



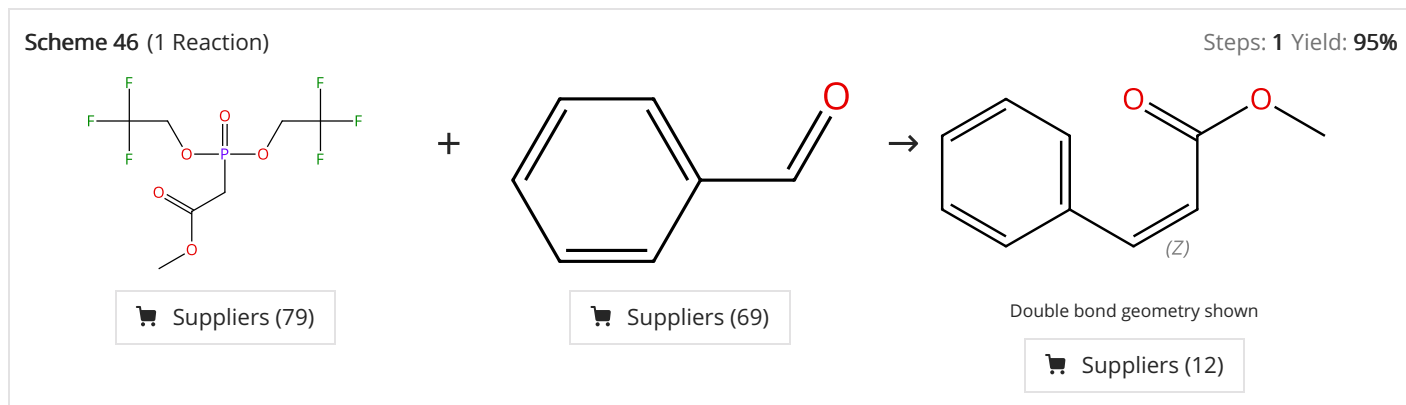
31-339-CAS-13137930 Steps: 1 Yield: 95% 1.1 Reagents: Sodium hydride Solvents: Tetrahydrofuran; 50 °C	Synthesis of pyrrolomorphinan derivatives as κ opioid agonists By: Fujii, Hideaki; et al Bioorganic & Medicinal Chemistry Letters (2010), 20(17), 5035-5038.
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Scheme 45 (1 Reaction)

Steps: 1 Yield: 95%



31-339-CAS-15267883	Steps: 1 Yield: 95% Synthesis of pyrrolomorphinan derivatives as κ opioid agonists
1.1 Reagents: Sodium hydride Solvents: Tetrahydrofuran; 50 °C	By: Fujii, Hideaki; et al Bioorganic & Medicinal Chemistry Letters (2010), 20(17), 5035-5038.



31-339-CAS-17555110	Steps: 1 Yield: 95% Potassium hexamethyldisilazide
1.1 Reagents: Potassium bis(trimethylsilyl)amide Catalysts: 18-Crown-6 Solvents: Tetrahydrofuran 1.2 -	By: Watson, Brett T. e-EROS Encyclopedia of Reagents for Organic Synthesis (2007), 1-21.